

LAHORE CITY, Pakistan

WMO: 416400

Lat: 31.550N

Lon: 74.333E

Elev: 215

StdP: 98.77

Time Zone: 5.00 (E05)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	6.8	7.9	0.2	3.9	17.4	1.8	4.4	16.6	4.7	16.6	4.0	16.5	0.1	310	0.347

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
6	9.6	42.3	23.4	40.9	23.1	39.4	23.5	29.0	34.1	28.6	33.7	28.2	33.4	1.7	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
27.9	24.6	32.0	27.4	23.8	31.7	26.9	23.2	31.6	97.1	33.7	94.9	34.2	93.0	33.7	34.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
5.2	4.1	3.4	DB	4.7	44.5	1.6	1.3	3.5	45.4	2.6	46.2	1.7	46.9	0.6	47.8	(4)
			WB	3.4	30.1	1.5	0.7	2.3	30.6	1.4	30.9	0.6	31.3	-0.5	31.8	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	25.2	13.4	17.1	22.4	28.5	32.8	33.4	31.6	31.0	29.9	26.5	20.6	15.1
	DBStd	7.38	2.26	2.74	3.44	3.27	2.89	3.17	2.27	1.96	1.99	2.48	2.05	2.44
	HDD10.0	4	3	0	0	0	0	0	0	0	0	0	0	1
	HDD18.3	318	152	53	7	0	0	0	0	0	0	0	3	102
	CDD10.0	5563	109	198	383	555	706	703	671	649	598	513	318	159
	CDD18.3	2836	0	18	131	305	448	453	413	391	348	254	72	2
	CDH23.3	39203	1	81	1127	4002	7127	7392	6201	5569	4662	2626	402	13
CDH26.7	23057	0	4	398	2247	4769	5061	3773	3154	2451	1149	51	0	
Wind	WSAvg	0.9	0.5	0.9	1.1	1.4	1.3	1.3	1.0	0.8	0.7	0.5	0.4	0.4
Precipitation	PrecAvg	618	22	27	37	19	20	54	188	156	72	9	5	11
	PrecMax	1118	121	118	167	141	111	209	478	556	450	106	63	112
	PrecMin	300	0	0	0	0	0	0	31	3	0	0	0	0
	PrecStd	209	25	28	34	23	22	51	101	113	69	16	10	19
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	22.6	27.2	34.6	40.9	43.9	44.5	40.4	37.3	36.7	35.1	29.7	24.8
		MCWB	14.7	17.6	21.2	21.1	22.4	23.6	26.7	27.1	25.6	24.0	18.8	15.4
	2%	DB	20.9	25.5	32.6	39.0	42.2	42.8	38.3	36.3	35.6	34.1	28.1	23.3
		MCWB	13.5	16.5	20.3	20.9	22.5	24.1	27.2	27.3	25.7	23.3	18.6	15.1
	5%	DB	19.7	24.1	30.6	37.3	40.9	41.3	37.1	35.5	34.8	33.2	27.0	21.8
		MCWB	12.9	15.8	19.8	20.5	22.3	24.2	27.6	27.3	25.8	22.9	18.0	14.2
	10%	DB	18.5	22.6	28.8	35.3	39.5	39.7	35.9	34.6	34.0	32.0	25.8	20.5
		MCWB	12.3	15.2	18.8	20.2	22.2	24.7	27.7	27.4	25.6	22.3	17.5	13.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	16.2	19.0	23.0	23.5	26.3	29.1	30.5	29.2	28.6	26.6	21.2	17.4
		MCDB	20.5	24.8	31.4	35.7	37.7	36.6	34.4	33.5	33.2	32.4	26.7	22.1
	2%	WB	15.1	18.0	21.8	22.5	25.2	28.2	29.2	28.7	27.9	25.8	20.2	16.4
		MCDB	19.1	23.6	29.8	34.5	37.1	35.6	34.2	33.1	32.8	31.3	25.6	21.3
	5%	WB	14.2	17.1	20.9	21.7	24.4	27.6	28.7	28.4	27.4	24.7	19.4	15.3
		MCDB	17.7	22.7	28.6	33.6	36.5	35.1	33.9	32.9	32.3	30.4	24.8	20.3
	10%	WB	13.2	16.1	19.9	21.1	23.7	27.0	28.3	28.1	27.0	23.7	18.6	14.3
		MCDB	17.1	21.1	27.3	33.0	35.9	34.8	33.6	32.7	31.9	30.2	24.1	19.1
Mean Daily Temperature Range	5% DB	MDBR	9.0	9.1	9.8	11.4	11.5	9.6	6.7	6.1	7.4	9.8	10.6	10.0
		MCDBR	11.1	10.5	11.5	13.5	12.8	11.8	8.0	7.1	8.0	10.1	11.5	12.0
		MCWBR	5.3	4.5	3.8	3.7	3.8	3.5	2.6	2.5	2.6	2.8	4.3	5.5
	5% WB	MCDBR	9.1	9.3	10.5	11.9	11.3	8.3	6.9	6.3	6.8	8.5	9.6	10.4
		MCWBR	5.0	4.4	4.1	4.1	4.4	3.6	3.2	2.7	2.8	3.1	4.3	5.4
Clear-Sky Solar Irradiance	taub	0.488	0.495	0.513	0.574	0.653	0.748	0.790	0.699	0.608	0.659	0.623	0.537	
	taud	1.855	1.889	1.879	1.788	1.653	1.527	1.498	1.653	1.784	1.596	1.614	1.755	
	Ebn at Noon	695	746	771	742	688	623	594	643	683	594	563	615	
	Edh at Noon	168	178	193	219	253	286	293	248	210	237	214	178	
All-Sky Solar Radiation	RadAvg	2.86	3.93	5.25	6.28	6.93	6.46	5.50	5.29	5.20	4.59	3.53	2.94	
	RadStd	0.33	0.41	0.33	0.28	0.24	0.37	0.35	0.41	0.28	0.22	0.19	0.18	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (5 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page